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Real-Time Applications in Autonomous Casing Running

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Abstract

This paper explores how real-time applications and autonomous control technologies are transforming casing running operations, addressing critical safety challenges in red-zone personnel exposure and operational inefficiencies. Traditional manual casing operations place workers in hazardous conditions with independently controlled traveling equipment, resulting in frequent injuries ranging from minor cuts to major accidents. The complex operational atmosphere on rig floors creates dangerous cycles where both personnel and equipment face elevated risks.

The smart tubular running system integrates comprehensive real-time sensor feedback, wireless control, and autonomous decision-making to eliminate red-zone exposure while improving operational efficiency. The system incorporates pressure, position, and temperature sensors across all components—including the Hydraulic Casing Running Tool (CRT), Flush Mount Spider (FMS), Link Tilt System, and Dynamic Bails—with data transmitted wirelessly to a smart control center for real-time analysis and display. Advanced interlock logic ensures proper conditions are met before tool activation, preventing misfires and unsafe operations. Virtual Thread Rep (VTR) technology enables automated torque-turn decisions with global remote oversight, allowing subject matter experts to intervene remotely during critical connections.

Field trials across North American land markets demonstrated significant performance improvements. Slip-to-slip time was reduced from 4.3 minutes to 2.6 minutes per joint when running 7-inch casing—a 30% improvement validated across six consecutive runs totalling 1,142 joints. The system achieved sustained performance of 23 joints per hour compared to 16 joints per hour with conventional methods. Safety enhancements included elimination of manual backup tongs and red-zone activities through autonomous sequencing and sensor validation.

The technology successfully completed Design Validation Testing per API Specifications 8C and 7K, including extensive laboratory testing and full-scale land rig trials. Post-deployment analysis showed improved operator confidence, reduced personnel requirements, and enhanced regulatory compliance through second-by-second operational logging. The system's modular design enables integration with existing hydraulic and mechanical CRTs while supporting diverse rig environments.

This autonomous, sensor-driven platform addresses key industry challenges by combining intelligent automation with data transparency, positioning smart tubular running as a transformative technology for

optimizing casing operations across complex global projects while advancing Industry 4.0 principles in drilling operations.

Keywords: Autonomous Casing Running, Real-Time Sensor Data, Smart Tubular Running, Real-Time Operations, Virtual Thread Rep, Red-Zone Safety, Torque-Turn Automation, Remote Oversight

Introduction

The oil and gas industry faces mounting pressure to improve operational safety while maintaining efficiency in casing running operations. Manual casing running inherently places personnel in hazardous "red zone" conditions on drill floors, where workers operate in direct proximity to powerful, independently controlled traveling equipment. This operational environment contributes to a concerning frequency of injuries, from minor incidents to major accidents, highlighting an urgent need for technological intervention. (De Mul 2020)

The complex operational atmosphere on rig floors introduces critical safety concerns. High noise levels and distracting environments significantly increase the potential for severe accidents. Misalignments, misfires, and inconsistent makeup further exacerbate these risks, creating dangerous cycles where both personnel and equipment face elevated threats. When equipment fails to function correctly and safety protocols are not consistently followed, problems multiply, leading to operational inefficiencies and substantial threats to personnel safety and asset integrity.

Current industry initiatives globally focus on removing personnel from the "red zone" of hazards and minimizing human exposure to high-risk operational zones. Regulatory and market-driven pressures increasingly mandate reduced personnel presence in critical red zones, reflecting a clear industry consensus on adopting safer, hands-free operational practices. However, despite clear safety advantages, a fundamental challenge persists: remote operations typically exhibit lower operational efficiency compared to traditional manual processes, creating reluctance in widespread adoption.

Existing casing running systems often lack comprehensive real-time visibility, automation capabilities, and traceable performance data. Traditional systems rely on manual methods and backup tools, exposing personnel to hazardous conditions while lacking integrated control. Beyond immediate rig floor operations, stakeholders have limited visibility into ongoing job progress, relying heavily on post-job summary reports. Real-time KPIs and detailed operational monitoring remain unavailable remotely, limiting the ability of stakeholders to effectively monitor, manage, and optimize performance. This inability to capture and leverage real-time operational data diminishes efficiency and complicates future planning efforts.

This paper presents the smart tubular running system as a comprehensive solution that integrates real-time sensor technology, autonomous control logic, and remote monitoring capabilities to address these challenges. The system demonstrates how advanced automation can enhance safety through red-zone elimination while maintaining or exceeding conventional operational efficiency. By combining intelligent automation with data transparency, this technology provides a pathway for industry-wide adoption of safer operational practices while supporting the energy sector's growing demand for operational excellence.

The Smart Tubular Running System Overview

The smart tubular running system encompasses a modular design with core elements; Hydraulic CRT, FMS, Link Tilt, Dynamic Bails, autonomous torque-turn analysis, and smart control center. The smart tubular running system is outfitted with various sensors including load, pressure, position, and temperature sensors giving the user real time feedback of the system status. The smart tubular running system also incorporates a new generation torque-turn sub which detects torque, rotational speed, number of turns, circulating pressure, and hook load. (Warren 2006)

The Hydraulic CRT features a modular design where the lower section can be installed depending on the casing size, weight and hook load requirements. The lower section can grip the casing internally or externally. The Hydraulic CRT is outfitted with temperature sensors which monitors the temperature of the internal bearings to detect and prevent premature wear or damage. It is also outfitted with position sensors which detect the position of the slips to ensure proper grip on the casing. Integral to the CRT are sensors that detects torque, rotational speed, number of turns, circulating pressure, and hook load. These internal sensors eliminate the requirement for an external Torque Turn Sub as well as providing key operational status of the tool while enabling the operational feedback required for autonomous activation.

The Link Tilt System with Dynamic Bails enables operators to efficiently manage single, double, or triple joints directly from the catwalk. When joints are positioned vertically, the link tilt system floats to center while the dynamic bails hydraulically retract, allowing seamless stabbing into the lower gripping assembly. This sequence significantly reduces operation time, driller complexity and provides a measurable improvement in repeatability. The FMS is outfitted with sensors which detect the position of the slips and set pressure to ensure proper grip and set position on the casing. The FMS provides enough grip force to hold backup torque on the first joint which eliminates the need for a manual backup tong.

The entire system is controlled by one wireless belly pack which allows the operator to remain in a safe area away from the red zone.

All data from the integrated system is wirelessly transmitted to the smart control center where it is analysed and displayed on the animated graphical display located on the rig floor, in the drillers cabin and remotely via a secure portal for offsite monitoring.

Interlock Logic and Autonomous Safety

The Hydraulic CRT and FMS are equipped with redundant interlock systems to ensure proper conditions are met before releasing. Before the CRT can be released logic conditions have to be met; FMS slip sensor detects proper slip position, FMS pressure sensor detects a minimum set pressure. Before the FMS can be released, conditions have to be met; Hydraulic CRT slip sensor detects proper slip position, Hydraulic CRT pressure sensor detects a minimum set pressure, ensuring the handoff sequence is in a safe state. This data is analyzed within the smart control center to verify all conditions are met before allowing function of the tool. The animated graphical display shows the condition of each component in the system giving the operator and driller real time tool status.

The Hydraulic CRT is also equipped with a mechanical interlocked bump plate. Once the Hydraulic CRT is stabbed into the casing and an appropriate weight is applied to the bump plate, the retainer locks will release and allow the tool to set in the proper position on the casing, preventing a common failure mode inherent to conventional tools.

Virtual Thread Rep (VTR) and Remote Oversight

The Virtual Thread Rep (VTR) system is an advanced technological solution designed to deliver real-time control and autonomous analysis of torque-turn operations. On-site torque-turn data is continuously collected by sensors and processed by an edge computing device utilizing decision-making rules provided by the thread Original Equipment Manufacturer (OEM). These rules are further customized according to regional training data and historical user applications. The processed data is securely transmitted to a secure cloud database and visually represented through an accessible online portal compatible with standard web browsers. Remote operators or experts can review these real-time visualizations, oversee operations, or intervene directly by making critical connection acceptance decisions remotely. (Powers 2011, Day 1990)

The versatility of the VTR system allows its deployment across diverse operational scenarios, including offshore, deepwater, and high-risk environments. Currently, VTR is being actively deployed in deepwater markets, facilitating real-time collaboration and oversight by National Oil Companies (NOCs), thread representatives, and key stakeholders. The system proved particularly indispensable during the COVID-19 pandemic, when personnel on-site was severely limited, and travel restrictions posed significant operational challenges. During this period, VTR successfully supported a series of multiple operations in North America, enabling torque-turn decision-makers located offsite to manage connection decisions remotely and effectively in real-time, ensuring uninterrupted and safe operations.

Traditional torque-turn services heavily depend on human judgment, which is vulnerable to fatigue, harsh operational conditions, and extended work hours, increasing the risk of human error. Any oversight or misinterpretation in connection analysis can have severe repercussions, as errors typically remain undetected until after well completion, potentially compromising wellbore integrity. VTR significantly mitigates these risks by providing autonomous oversight and immediate validation of connection quality, reducing the likelihood of undetected errors. Furthermore, the ability to remotely review connection decisions simultaneously from various locations serves as an additional protective measure, enhancing decision-making reliability and ensuring high-quality, secure well construction.

Testing, Validation, and Deployment

The Design Validation Testing (DVT) process for the smart tubular running system was rigorously conducted to ensure its functionality and compliance with industry standards. The system was designed and manufactured in accordance with API Specifications 8C and 7K, which are recognized benchmarks for equipment used in drilling operations. To validate the system's performance and reliability, it underwent extensive lab testing, where functional requirements such as tool positioning, torque control, and real-time data monitoring were thoroughly examined. These tests ensured that all critical aspects of the system met the required specifications before advancing to field trials. Additionally, the system was subjected to a full-scale test on a land rig, simulating real-world conditions in a controlled test environment. This pre-field trial validation allowed engineers to fine-tune the system and identify any potential issues, ensuring the technology was ready for deployment. (API 7K 2015, API 8C 2012)

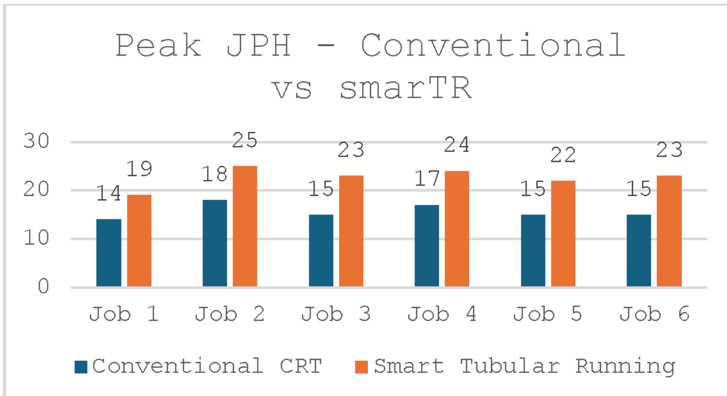
Once the design and lab testing phases were completed, the Field Trial Deployment ranging from select elements of the package to the complete smart tubular running system took place in key locations across the globe, providing a real-world test of its capabilities. These trial locations included North America land markets — regions known for their challenging and high-stakes drilling environments. In these trials, the system demonstrated significant improvements in operational efficiency, most notably a reduction in slip-to-slip time, which is a critical metric in drilling operations. By automating key tasks, the system also alleviated operator fatigue and reduced the overall headcount needed on-site, supporting safety initiatives that are designed to remove personnel from hazardous red zone activity. The field trials were instrumental in confirming the system's ability to enhance safety, increase consistency, and streamline the wellbore construction processes, providing valuable insights for further refinement before full-scale deployment.

As part of the comprehensive testing process, special attention was given to Cybersecurity and Software Reliability to ensure the system's resilience in the field. Given the reliance on real-time data transmission and remote monitoring, the smart tubular running system was designed with secure communication protocols to protect against cyber threats and unauthorized access. Firmware resilience was also a critical focus, ensuring the system could withstand hardware and software failures without compromising performance and reliability. The system’s security architecture was rigorously tested to meet industry standards, ensuring the integrity and confidentiality of operational data. These measures helped build confidence among stakeholders, knowing that the system was not only efficient and effective but also secure and reliable in demanding field environments.

Results and Observations from Field Trials

Field deployment of the smart tubular running system demonstrated measurable safety and efficiency improvements in diverse field environments. For example, on a recent US Land rig, the slip-to-slip time decreased from an average of 4.3 minutes per joint (conventional CRT makeup process) to 2.6 minutes per joint running 7" casing with the integrated system—a 30% reduction. This result was validated across six consecutive runs ranging from 7" to 7-5/8" with a cumulative sample size of 1,142 joints.

Table 1—Is a baseline study that compares operational efficiency between conventional CRT makeup (16 joints per hour average) and smart tubular running system (23 joints per hour average), demonstrating a 30% improvement in performance.



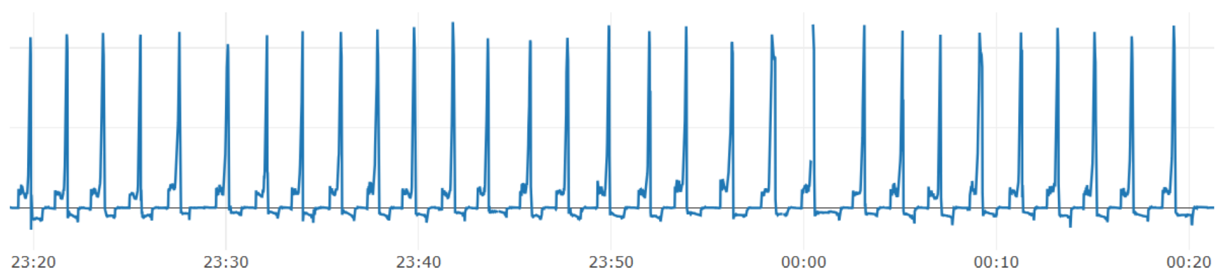


Figure 1—Represents final torque measurements rather than the complete torque profile during connection makeup. The data demonstrates improved performance. Recent field applications have achieved higher efficiency rates, surpassing the performance metrics shown in this baseline analysis.

The graphical driller interface enabled operators to instantly interpret tool status and act proactively, preventing misfires and reducing non-productive time.

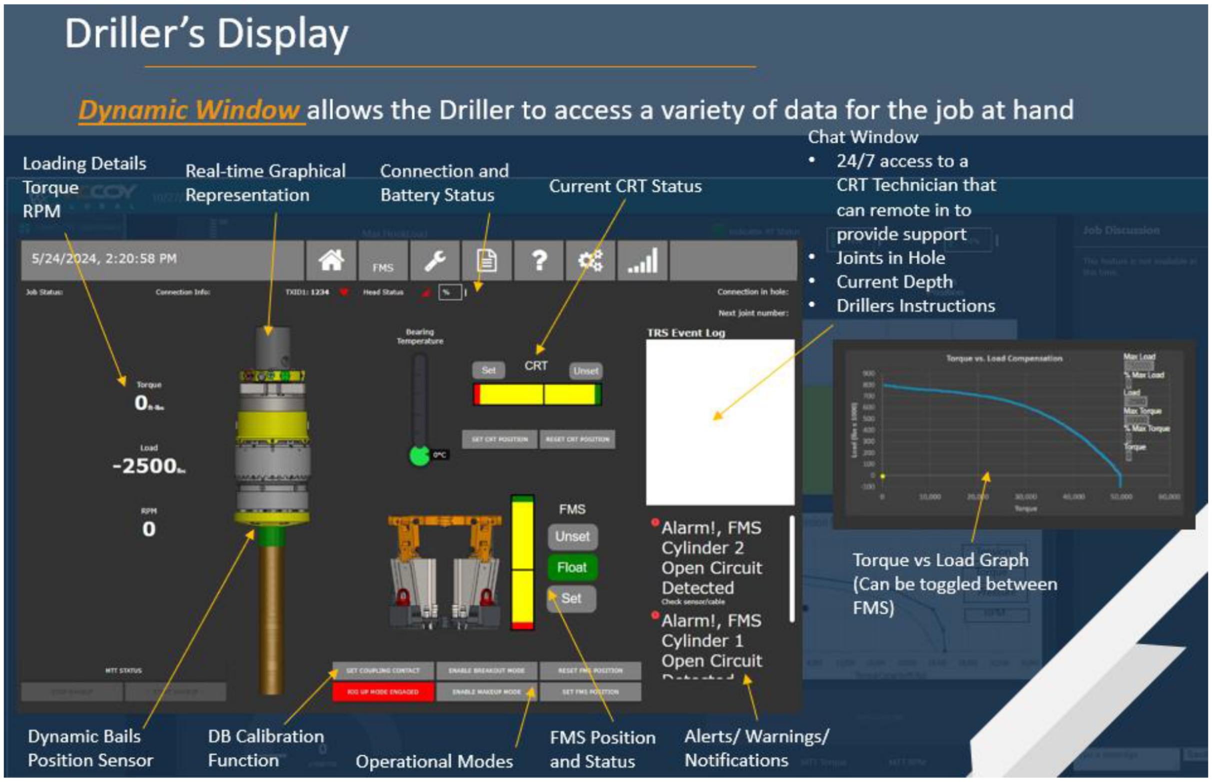


Figure 2—Displays the graphical user interface enabling operators to monitor tool engagement, slip positioning, and system status from outside the red zone.

Continuous data collection facilitated precise interlock validation, while live telemetry and remote sensing improved system alignment and enabled autonomous verification of tool activation. VTR enabled real-time torque-turn autonomy with remote oversight, allowing subject matter experts to intervene remotely during critical connections. As a result, repeatability and consistency across operations improved. Post-job, time-stamped reports delivered objective transparency, supporting KPI tracking, root cause analysis, and regulatory compliance. Notably, operator feedback cited increased confidence in both tool reliability and safety.

Together, these autonomous, sensor-driven capabilities address key industry challenges by combining intelligent automation with data transparency. Smart tubular running is positioned as a transformative platform for optimizing casing operations across complex global projects. Operators reported improved KPIs, stronger safety compliance, and reduced personnel requirements. Its modular design also enabled smooth integration with hydraulic and mechanical CRTs, minimizing non-productive time and supporting diverse rig environments.

One of the standout benefits of smart tubular running is the elimination of red-zone tasks. In traditional operations, certain critical tasks, often referred to as "red zone" activities, carry a high risk of failure or injury. By shifting these tasks to autonomous cycles, the system mitigates human error during these crucial stages. Operators can rest assured that the tool activation and positioning are being performed according to preset parameters, ensuring no misfires or unsafe operations occur. The result is a significant reduction in potential safety hazards and worker fatigue.

The system also generates continuous logs, which play a crucial role in post job analysis and audit readiness. Each operation is meticulously recorded, creating a complete, transparent timeline of actions and tool statuses. This granular data is essential for troubleshooting and continuous improvement, as operators and engineers can trace any anomalies to their exact origin. This level of transparency not only boosts operational efficiency but also enhances accountability and future decision-making, ensuring that any deviations are swiftly addressed with data-backed insights.

Strategic Differentiators

Traditional tubular running systems rely on manual methods and backup tools, which expose personnel to hazardous red-zone conditions and lack integrated control. In contrast, the smart tubular running system incorporates an FMS that reacts torque directly into the rotary table, eliminating the need for manual backup tongs. Sensor feedback ensures the FMS is properly set and gripping the casing before visually signalling to the driller that it is safe to apply torque. This significantly enhances operational safety and reliability. Additionally, while conventional systems operate without real-time sensor feedback, smart tubular running system offers continuous high-resolution data streams that support real-time decision-making and system diagnostics.

Conventional systems typically consist of independent, manually operated components requiring constant coordination between operators. The smart tubular running system differentiates itself with full system integration through components like the belly pack, VTR, smart control center, and robust interlock logic. These features allow for autonomous sequencing based on real-time sensor validation across the system, reducing human error and enhancing repeatability. The elimination of verbal or hand-signal-based communication also streamlines operations and minimizes miscommunication.

The smart tubular running system aligns with Industry 4.0 principles by enabling interoperable, digital, and autonomous operations. It supports communication with third-party systems through multiple industry-standard communication protocols, making it ready for broader integration with rig control systems. This connectivity enables the expansion of system interfaces to include rig-based equipment such as drilling recorders and traveling systems, creating a pathway to fully autonomous rig environments. With its advanced digital architecture, smart tubular running system lays the foundation for predictive analytics, data-driven decision-making, and the evolution of autonomous well construction workflows.

Future Roadmap and Expansion

The future development of the smart tubular running system will incorporate artificial intelligence and machine learning (AI/ML) to enhance anomaly detection, operational repeatability, and efficiency optimization. As performance data accumulates across deployments, AI models will be trained to analyze each element in the casing sequence—identifying inefficiencies, detecting outliers, and recommending real-time adjustments to sequence execution. This iterative, data-driven refinement is expected to yield measurable improvements in safety, consistency, and non-productive time reduction. (Krikor 2022)

Beyond tool-level automation, future development will include expanded system intelligence that drives guidance for sequence optimization and predictive diagnostics. This includes AI-informed control over operations such as pipe presentation at the catwalk, automated centring and stabbing motions, and monitoring casing travel rate through top drive data and surface pressure responses. For example, monitoring torque and hook load trends in parallel with set-down speed and calculated string weight will enable the system to detect unsafe running speeds or excessive drag, prompting the system to intervene or alert the driller before damage occurs. Similarly, precise integration with rig sensors will allow the system to measure and verify actual casing tally length, improving job planning and cement displacement accuracy beyond the conventional method of measuring and recording these critical measurements offline.

The smart tubular running system is already demonstrating its adaptability across diverse operational environments. It is currently deployed in deepwater markets with an integrated tong package interfacing with iron roughneck positioning systems. Although presently operated independently, future deployments aim to fully integrate these rig systems with smart tubular running to enable seamless, coordinated operation. Additionally, the system is actively used in unconventional drilling campaigns through smart casing running applications, proving its versatility in both high-volume and high-risk environments.

The long-term roadmap envisions smart tubular running as a foundational element in the autonomous rig ecosystem. In this vision, high-level commands from rig control platforms will be executed locally by the smart tubular running system. Integration with rig subsystems—including traveling equipment, pipe-handling arms, torque-turn monitoring, and drill floor sensors—will ensure coordinated sequencing, interference mitigation, and repeatable execution. The system’s open communication protocols, cloud connectivity, and sensor-driven architecture position it as a scalable, interoperable platform to support fully autonomous casing operations and broader rig digitalization initiatives.

Conclusion

Smart tubular running technology represents a fundamental advancement in wellbore construction, delivering enhanced safety through logic-driven interlock systems that remove personnel from hazardous red-zone operations. The system's real-time monitoring, autonomous control, and sensor-based validation capabilities improve operational efficiency while providing precise, repeatable performance.

This integrated platform benefits the entire oilfield service chain—from tubular running providers to drilling contractors and E&P companies—by reducing non-productive time and enabling data-driven decision-making. As digital rig architectures continue to evolve, smart tubular running systems are positioned as essential components of modern drilling operations.

To meet the performance, safety, and efficiency expectations of future well construction, the industry must accelerate the adoption of real-time, autonomous casing running technologies. The smart tubular running system provides a proven foundation for this transition and stands ready to support scalable deployment across global operations.

References

1. API Specification 7K (2015). Drilling and Well Servicing Equipment, 6th Edition
2. API Specification 8C (2012). Drilling and Production Hoisting Equipment, 5th Edition.
3. Day, J.B., Moyer, M.C., Hirshberg, A.J. (1990). New Makeup Method for API Connections. *SPE Drilling & Completion* **5**(3):233–240.
4. De Mul, A. (2020). Full Automated Casing Running: The Next Step Completed in Field Automation. ADIPEC (SPE).
5. Krikor, A., et al (2022). Machine Learning Delivers Automated Rig-State Detection and KPI Computation. ADIPEC (SPE). Paper SPE-211753-MS.
6. Powers, J.P., Chelf, M.S. (2011). Continuing Application of Torque-Position Assembly Technology for API Connections. *SPE Drilling & Completion* **26**(3):408–418.
7. Warren, T., et al (2006). Top-Drive Casing-Running Process Improves Safety and Efficiency. *SPE Drilling & Completion*.